



ST BENEDICT'S COLLEGE

SUBJECT	LIFE SCIENCES PAPER 1	DATE	21 JULY 2023
GRADE	11	MARKS	150
EXAMINER	MRS S. STEGMANN	MODERATOR	MRS B. ZAJAC
		DURATION	2 ½ HOURS

MARKING GUIDELINES

Question 1:

- 1.1. D ✓
H ✓
F ✓
K ✓
C ✓
E ✓
B ✓
I ✓
G ✓
L ✓

[10]

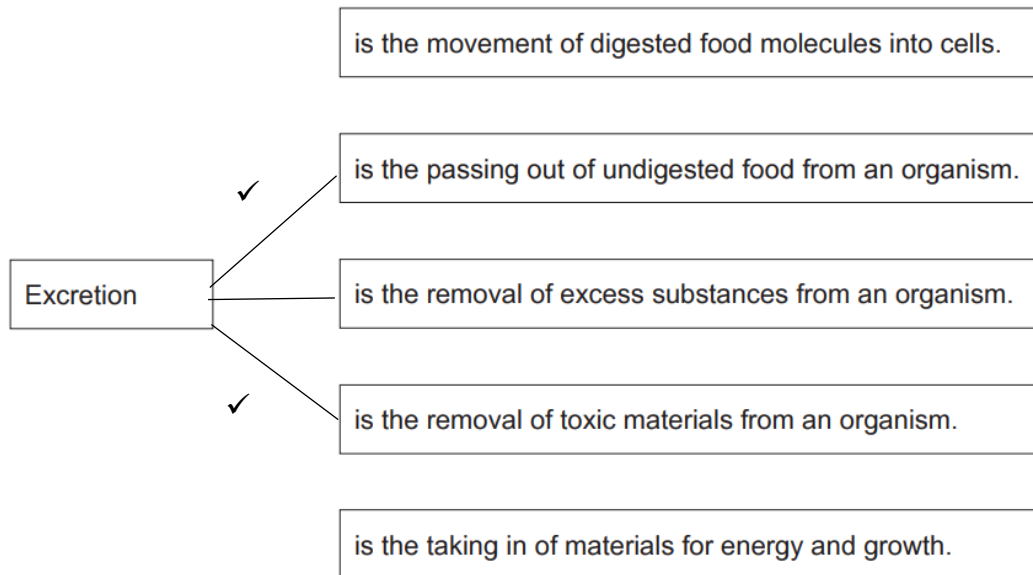
- 1.2. Various possibilities are suggested as answers to the following questions. Indicate the correct answer by writing the correct letter from the corresponding question in the space provided in the table below:

1.2.1.	1.2.2.	1.2.3.	1.2.4.	1.2.5.	1.2.6.	1.2.7.	1.2.8.	1.2.9.
C ✓	D ✓	A ✓	B ✓	A ✓	B ✓	B ✓	C ✓	A ✓✓

[10]

- 1.3.
- 1.3.1. Disease-causing organism/ agent ✓ (1)
- 1.3.2. A ✓ (1)
- 1.3.3. $8200 / 248\,490 \times 100 = 3,29\%$ ✓ adults (accept if rounded down to 3) (3)
- 1.3.4. Make people more aware about how HIV is transmitted
- Educate people on the STI
- Make condoms more readily available
- Any acceptable answer ✓✓ (2)
- 1.3.5. Heading (H) ✓✓
- X axis (title + scale)- (X) ✓
- Y axis (title + scale)- (Y) ✓
- Bars same width + spacing the same (B) ✓
- Plotting (P)✓✓ (7)

[14]



(any2)

1.4.2. Lungs- CO₂

Large intestine/ colon- faeces

Skin- sweat ✓✓✓✓

(any 2 x 2=4)

1.4.3. Water

Urea

Uric acid

Creatinine

Salts ✓✓✓

(any 3)

1.4.4.

A. P- renal artery ✓

Q- Bladder ✓

R- Ureter ✓

(3)

B. Table showing the difference in blood between vessel A and B/ inferior vena cava and aorta ✓

Differences:

- A carries blood high in CO₂ and B carries blood low in CO₂
- A carries blood low in O₂ and B carries blood high in O₂
- A carries purified blood and B carries unfiltered blood

any 2 x 2 ✓✓✓✓

-1 Table incomplete

(5)

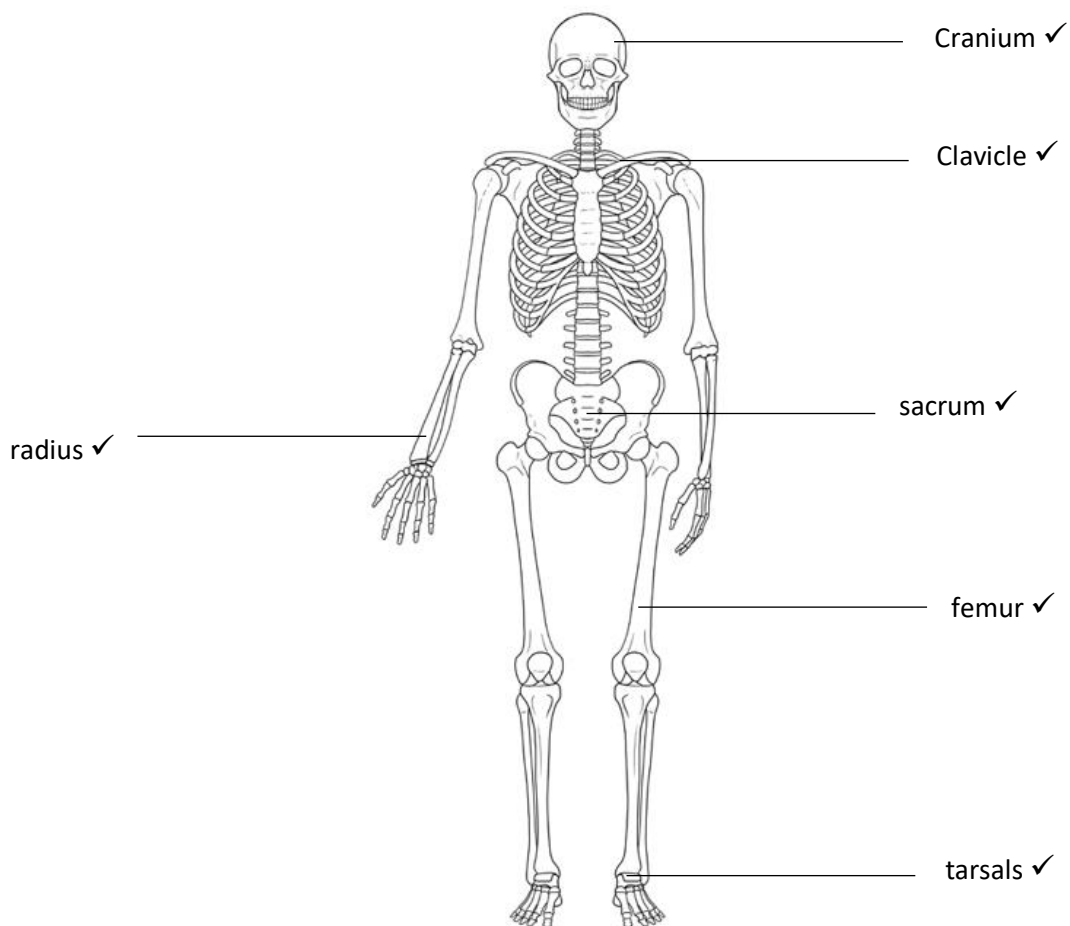
1.4.5.

Condition	Volume of urine	Concentration of urine
Increase in water consumption	Increases	Decreases ✓ (both correct)
Increase in temperature	Decreases	Increases ✓ (both correct)
Increase in exercise level	Decreases	Increases ✓ (both correct)

(3)

[20]

1.5.



[6]

Total Question 1: 60 marks

Question 2

- 2.1.
- 2.1.1. The removal of harmful metabolic wastes from the body✓✓ (2)
- 2.1.2. Renal corpuscle ✓ (1)
- 2.1.3. The dialysis fluid contains no waste substances✓/ no nitrogenous wastes/urea, uric acid, etc. whereas the patient's blood does. ✓
The glucose, amino acids, salts and other blood constituents are in the same concentration in both the dialysis fluid and patient's blood ✓✓ the fluid has a similar composition to plasma (4)
- 2.1.4. A concentration gradient needs to be set up and maintained between the patient's blood and dialysis fluid for waste substances✓ only so that these diffuse out of the blood into the dialysis solution. ✓ (2)
- 2.1.5. As waste substances move into the fluid the concentration gradient becomes less and less. ✓
It is replaced to maintain the concentration gradient and ensure maximum diffusion of waste out of the blood. ✓ (2)
1 mark for sterilization- sterile fluid.
- 2.1.6. Creates a large surface area✓ to ensure maximum absorption of waste✓ (2)
- 2.1.7. Time consuming
Very expensive
Hard to hold down a job
Have to limit fluid intake
Potassium and phosphates are not removed and can build up and become life-threatening
May feel ill during treatment
Anticoagulants are added which may lead to bleeding problems. ✓✓ (any 2)
- 2.1.8. Kidney transplant ✓ (1)
- [16]

- 2.2.
- 2.2.1. Nephron ✓ (1)
- 2.2.2. Renal vein ✓ (accept inferior vena cava) (1)
- 2.2.3. Glomerulus ✓ (1)
- 2.2.4. 1- cortex ✓
3- Medulla ✓ (2)
- 2.2.5. Proximal convoluted tubule ✓ (1)
- 2.2.6. Most useful substances are reabsorbed by active transport✓ in the PCT. This requires a lot of energy ✓and therefore a lot of mitochondria (2)
- 2.2.7.
- a. Protein is too big to be filtered out of the glomerulus and will therefore not be present in the filtrate (region 1) but will remain in the blood plasma ✓✓ (2)
- b. Glucose is filtered out of the blood and is present in the filtrate (region 1) ✓ it is reabsorbed completely in the PCT ✓and will therefore not be present in the Loop of Henle (region 3) (2)
- c. Urea is actively secreted ✓ back into the kidney tubules from the blood ✓so region 3 will contain higher amounts (2)

[14]

Total Question 2: 30 marks

Question 3

- 3.1.
- 3.1.1. B- radium ✓
C- ulna ✓
D- humerus ✓ (3)
- 3.1.2. Tendon ✓
Attaches muscle to bone to bring about movement ✓ (2)
- 3.1.3. Surgery to repair and reattach the tendon ✓
Then a cast/sling to immobilize the joint ✓ (2)
- 3.1.4.

A table comparing various characteristics of two synovial joints ✓

	Name of synovial joint	Type of movement allowed	Location
1	Ball and socket ✓	Most flexible- allows for movement in many directions ✓	Hip/ between femur and pelvis ✓
2	Hinge ✓	Allows for movement up and down ✓ (NOT side to side)	e.g. phalanges, knee, ankle ✓

- 3.1.5.
- A. Rickets ✓
B. Osteoporosis ✓
C. Arthritis/ rheumatoid arthritis ✓ (3)
- [17]

- 3.2.
- 3.2.1. a chronic, inflammatory disorder in which your immune system mistakenly attacks the lining of membrane that surrounds your joints. ✓✓ (2)
- 3.2.2. The ligaments and surrounding soft tissue are affected ✓ and the joints weaken. ✓ Deformities then occur. (2)
- 3.2.3. Synovial fluid ✓ (1)
- 3.2.4. 98% of RA patients experienced foot pain whereas only 76% of patients without arthritis had foot pain ✓
96% of RA patients reported difficulty in function and 66% without arthritis had function difficulties ✓ (2)

3.2.5. Same gender

Same age group

Same length of time having RA

Same kind of RA

Same number of patients in each group ✓✓

(any 2)

3.2.6. An enlarged bursa at the base of the big toe✓. The joint becomes thickened, bent and often inflamed. ✓ OR

A bony bump that forms on the joint at the base of your big toe. ✓ Occurs when some of the bones in the front part of your toe move out of place✓ (2)

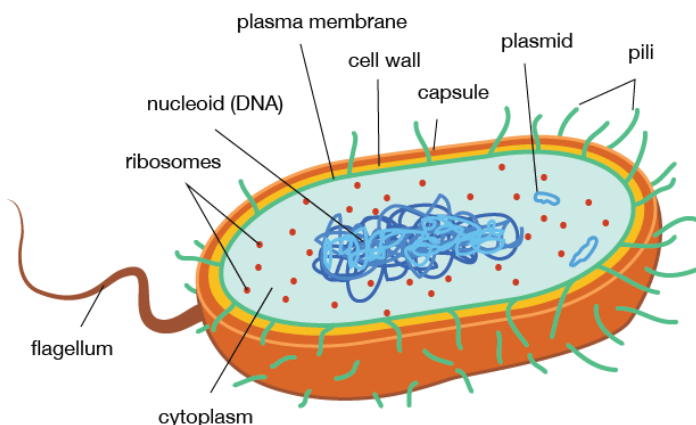
3.2.7. Cartilage that cushions the ends of bones in joints softens and wears away✓. Bones rub against one another causing pain and stiffness. ✓ (2)

[13]

Total Question 3: 30 marks

Question 4

- 4.1.
- 4.1.1. Alicin ✓ (1)
- 4.1.2.
- a. Contents of the test tube ✓
 - b. Diameter/ growth of bacteria colony ✓
 - c. same amount of milk; same time/period to do the investigation; same environmental conditions/temperature ✓✓ (any 2) (4)
- 4.1.3. To avoid the growth of bacteria✓ before the start of the experiment as most bacteria do not grow in cold conditions. ✓ (2)
- 4.1.4. Petri dish A: there was very little bacterial growth✓ as no E.coli was added to the sample. The already existing bacteria on the agar is the growth that is seen. ✓ (2)
- Petri dish C: The E.coli specimen and garlic extract did not show any signs of bacterial growth. ✓
- The alicin/ antimicrobial substance in the garlic extract destroyed the bacteria. ✓ Hence there was no growth in C. (2)
- 4.1.5. It is a medium/ source of nutrition for microorganisms ✓ (1)
- 4.1.6. No. ✓
- Antibiotics are only effective against bacteria. Garlic is partially effective against bacteria and viruses. ✓ (2)
- 4.1.7.



- Heading ✓
- Accuracy and diagram rules ✓
- Any 3 labels ✓✓✓ (5)

- 4.2.
- 4.2.1. A relationship/ association between two organisms, where both benefit. ✓ (1)
- 4.2.2. The hyphae of the fungus increase the absorptive surface of the plant's roots ✓/ allow the plant to absorb water, phosphorous and other minerals from the soil.
The plant provides the fungus with carbohydrates ✓ (2)
- 4.2.3. Same type of soil was used
All other factors were kept the same ✓ (any 1)
- 4.2.4. Sterilizing the soil killed the fungi ✓ meaning that
there were no mycorrhizal hyphae ✓
to increase the absorptive surface of the plant's roots. ✓
The plant could not get sufficient water, phosphorous and other mineral ions from the soil, ✓
resulting in slow growth. (any 3)
- 4.2.5. Mycorrhizal fungi encourage plant growth as can be seen in plants grown in non-sterilised soil.
✓✓ (2)
- 4.2.6. Parasitism- one organism benefits and the other is harmed. ✓✓ OR
Commensalism- one organism benefits and the other is unchanged/ unaffected ✓✓ (2)

[11]

Total Question 4: 30 marks

TOTAL: 150 MARKS